

Having a machine-learning algorithm that analyses the data based on all the feature vectors determined will produce an objective ranking with hard history on the way the ranking was arrived at. This avoids any subjective judgment in such critical tasks. On the other hand, consider a fuzzy problem such as whether a conversation between a group of people contains racial undertones. Human intuition driven pattern detection (built over centuries of our evolution) can detect undertones of the conversation and classify it more accurately than any machine-learned algorithm. Accurate discovery based on such approximate impressions needs interactive local analytics on raw *data at rest*.

When the classification feature vector is of a huge dimension and the context/scenario is fuzzy, human driven contextual local analytics is more successful. This, in turn, necessitates that the data should be presented in a manner conducive for human driven pattern recognition to act on it. We propose building smart storage for IoT devices capable of approximate shallow analytics tasks. Since IoT devices are thin, low power edge devices, such devices are capable only of running energy aware shallow analytics in-situ. Decentralised in-situ local analytics driven by human interaction at such smart IoT devices needs to be built into the IoT storage. Such edge devices will participate in hierarchical insight discovery wherein local insights relevant to the data generating entity are performed at the origin,

whereas insights gleaned from the long tailed data can be done at the back-end cloud.

In-situ visualisation is another key need for the analysis of *data at rest*. Human beings are great at detecting patterns. Insight discovery analytics needs to run locally on the sensor device in a way that is interactive with humans. Visualisation is not about fancy pie charts, but about making sense of data and identifying patterns, with human-aided machine learning. We will discuss more on in-situ visualisation for *data at rest* in next month's column.

If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Till we meet again next month, happy programming! 

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